

Course: Software Design Pattern

Code: MITE 435

Semester: Fall 2026

Course Description

This course introduces students to design patterns as reusable solutions to common problems in object-oriented software design. Students will learn the theory behind Creational, Structural, and Behavioral patterns, understand when and why to apply them, and practice implementing them through hands-on labs and a capstone project. Emphasis is placed on writing maintainable, extensible, and testable code using SOLID principles alongside the Gang of Four (GoF) patterns.

Assessment Structure (up to mid)

Component	Weight
Attendance & Participation	5%
Lab Assignments	20%
Midterm Exam	25%

Weekly Breakdown

Week 1: Introduction & Foundations Introduces why design patterns matter, their origin in the GoF book, and key design principles (SOLID, DRY, KISS, YAGNI). Ends with a UML refresher on class and sequence diagrams. Refactoring legacy code with patterns.

Week 2: Introduction to Design Pattern Categories: Basic OOP behaviours: coupling, cohesion, and abstraction. Overview of Creational, Structural, and Behavioral patterns. Introduces anti-patterns and code smells to spot poor design.

Week 3: Creational Patterns I: Singleton (single shared instance) and Factory Method (delegating object creation to subclasses).

Week 4: Creational Patterns II: Abstract Factory (families of related objects), Builder (step-by-step object construction), and Prototype (cloning objects).

Week 5: Structural Patterns I: Adapter (bridging incompatible interfaces), Bridge (decoupling abstraction from implementation), and Composite (treating individual and grouped objects uniformly).

Week 6: Structural Patterns II: Decorator (adding behavior dynamically), Facade (simplifying complex subsystems), and Proxy (controlling access to objects).

Week 7: Midterm Exam: Covering design principles, Refactoring, Creational and Structural patterns, and pattern selection based on UML/scenarios.